

Computer Vision (CSE344)

Final Project Presentation

Polyp Segmentation, Tracking & VLM Consultation in Colonoscopy

Ishaan Shekhar [2023249], Krish Aggarwal [2023286], Harshit Anand [2023236], Abhas Gupta [2023017]
Engg. Track



INDRAPRASTHA INSTITUTE *of*
INFORMATION TECHNOLOGY
DELHI



Problem Statement, Scope & Users



Manual Colonoscopies have high rates of polyp misses, while patients usually lack transparency and knowledge regarding their medical procedure outcomes. In order to solve this issue, we are working on a prototype of a web-based application that will include **pixel-wise segmentation, continuous object tracking, and vision-language model**. The system will provide video stream with segmentation along with dynamically tracked polyps. Users will be able to ask questions about specific polyps on our VLM interface.

Our product targets two main user personas- **gastroenterologists** who use clinical insights to assist in post-operative reporting and decision-making, and **patients**, who can use the VLM interface to ask questions, get second opinions and translate complex metadata into understandable information.

Other Potential Users:

- **Gastroenterology students and trainees:** They can use the system as an educational tool on archived videos.
- **Pathologists:** Better localization can improve biopsy targeting and histopathological correlation.
- **Medical imaging and AI researchers:** Can use this project as a baseline and potentially expand it.

Related Work & Baseline Methods



Ghatwary et al. (2026) present a multi-center analysis of deep learning methods for video-based polyp detection and segmentation using the EndoCV2022 dataset. While the paper shows the benefits of temporal modeling, it mainly captures short-term dependencies and lacks long-term temporal reasoning. It does not include explicit object-level tracking or unique polyp identification, limiting clinical usability for counting and monitoring. Additionally, it focuses only on detection and segmentation without higher-level reasoning or explainability, and lacks structured tracking evaluation. These gaps motivate our approach integrating tracking, unique polyp identification, and vision-language reasoning.

PolypSegTrack introduces a unified transformer-based framework that jointly performs polyp detection, segmentation, classification, and tracking using shared object-query representations and a conditional mask loss for flexible training. It employs query-based matching across consecutive frames for unsupervised tracking. However, the method lacks explicit temporal modeling as it relies on pairwise frame matching rather than learning long-term motion dynamics. Additionally, its transformer-heavy design leads to high computational cost, and the reliance on natural image pretraining without domain adaptation may limit performance in medical-specific scenarios.

Self-Prompting Polyp Segmentation in Colonoscopy Using Hybrid YOLO-SAM 2 Model: The YOLO-SAM2 model proposes a hybrid pipeline where YOLOv8 generates bounding box prompts that are fed into SAM2 for precise segmentation, eliminating the need for manual prompts and reducing annotation effort. It leverages SAM2's zero-shot capability and memory mechanism to enable accurate and real-time segmentation on video data. It's 44 FPS processing allows for real time deployment. However, the method is heavily dependent on detection quality, as segmentation fails when YOLO misses objects, leading to error propagation. Additionally, it lacks identity-aware tracking limiting its ability to distinguish unique polyps across frames.

EndoChat is a multimodal Vision-Language Model designed for endoscopic surgery that maps visual features to text, enabling users to ask open-ended clinical questions about surgical frames.

Related Work & Baseline Methods



Its end-to-end architecture relies entirely on visual estimation, making it prone to hallucinating quantitative metrics like polyp sizes. Our project will improve upon this limitation by separating detection and reasoning procedures. We extract precise mathematical metadata (polyp IDs and segmentation areas) and feed it directly into the VLM, ensuring our consultation tool provides strictly grounded, reliable clinical answers.

Polyp-PVT: Polyp Segmentation with Pyramid Vision Transformers by **Dong et al. (2023)** introduces a transformer-based segmentation framework leveraging Pyramid Vision Transformers for hierarchical multi-scale feature representation and long-range dependency modeling. In contrast to prior CNN-based methods with limited receptive fields, the proposed architecture integrates Cascaded Fusion Modules, Camouflage Identification Modules, and Similarity Aggregation Modules for effective cross-level feature fusion and boundary refinement. This design enables superior delineation of small, low-contrast, and camouflaged polyps, achieving improved performance over previous state-of-the-art approaches.

EndoVLA (Ng et al., CoRL 2025) is a two-stage Vision-Language-Action model, based on Qwen2-VL-7B with LoRA and DFT (SFT + RFT on a ~6k phantom dataset), which takes an endoscopic RGB image and a language instruction as input and generates a bounding box and a robot bend action for autonomous polyp tracking. Although EndoVLA is pertinent to the task of polyp detection and tracking in colonoscopy videos, the limitations of the VLA module include heavy computation per frame (operating at only ~2 Hz while receiving 30 FPS video), lack of temporal information for each frame, no fine-grained object segmentation, and support only for one target and a single discrete action.

Baseline Methods :

Object Detection : **YOLOv8**

Segmentation : **Polyp-PVT [Dong et al. 2023]**

Datasets & Evaluation Metrics



Kvasir-SEG Dataset

A medical imaging dataset containing 1,000 high-resolution colonoscopy images with expert-annotated pixel-wise masks for training and evaluating models for automated colorectal polyp segmentation.

CVC ClinicDB

CVC-ClinicDB is a database of frames extracted from colonoscopy videos. The dataset contains several examples of polyp frames & corresponding ground truth for them. The Ground Truth images consists of a mask corresponding to the region covered by the polyp in the image.

PolypGen2.0

A multi-center colonoscopy dataset with ~3,300 annotated frames (from 6 hospitals), containing both polyp and non-polyp images with pixel-level masks and bounding boxes. It is designed to test real-world generalization, with diverse conditions (blur, occlusion, varying polyp sizes) and a separate multi-center test set to evaluate model robustness.

The system is evaluated across segmentation, detection, tracking, and reasoning components.

Segmentation: Dice Coefficient and IoU measure overlap between predicted and ground-truth polyp regions. Also, pixel-level precision and recall are used to assess segmentation quality.

Detection: AP, Recall (Sensitivity), and mAP(0.5–0.95) evaluate detection accuracy, ensuring minimal false positives and missed polyps.

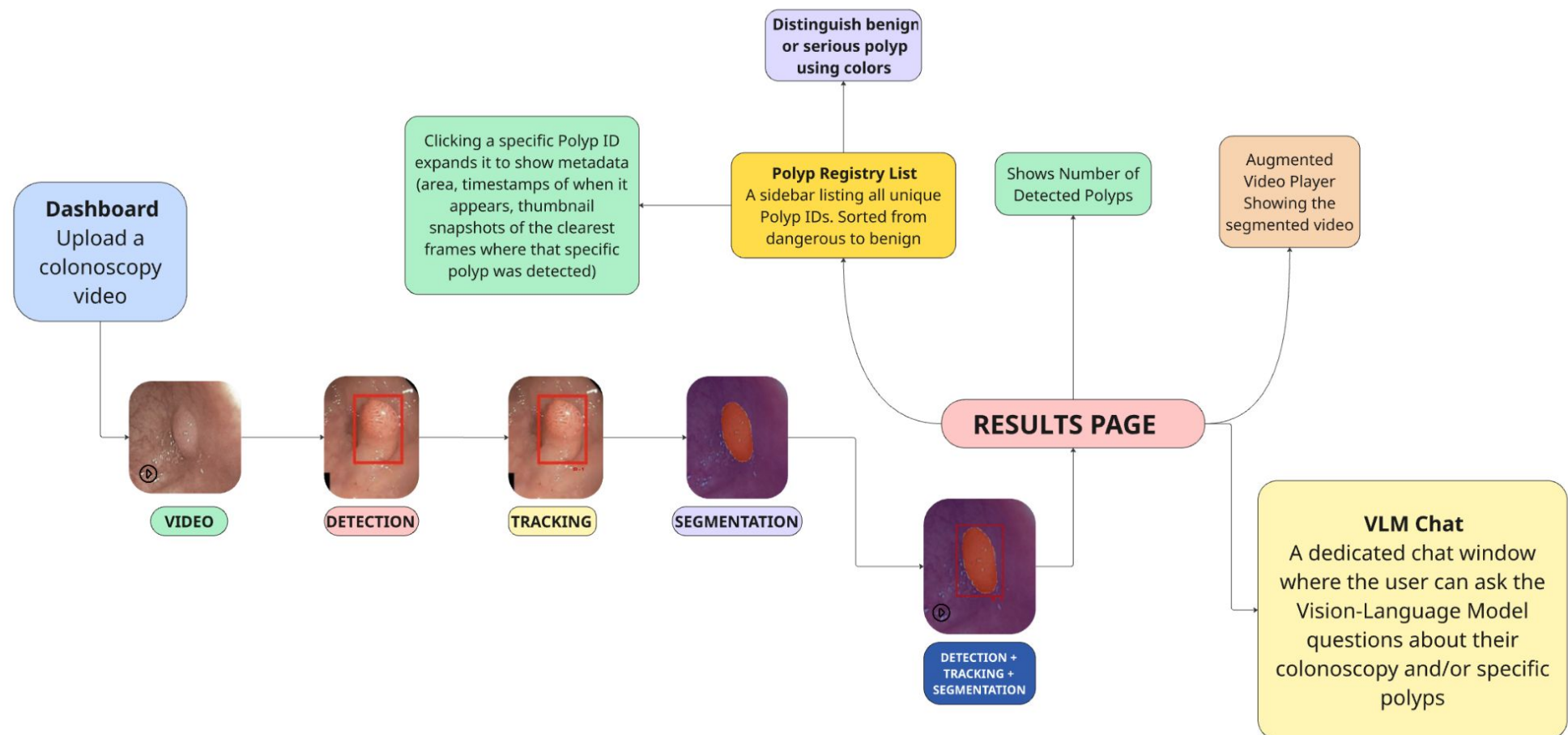
Tracking: In the absence of ground-truth tracking annotations, temporal behavior is evaluated using frame-to-frame IoU to measure the stability of predicted polyp regions across consecutive frames. Higher IoU indicates consistent localization over time. Additionally, qualitative inspection is performed to assess temporal coherence and consistency of detections.

Vision-Language QA: Answer correctness and semantic relevance, and grounding with respect to visual inputs and extracted metadata.

System



[Miro link](#)



Analysis of Baseline Results



```
mDice      : 0.8232
mIoU       : 0.7192
F_beta^w   : 0.8600
S_alpha    : 0.9526
mE_xi      : 0.9460
maxE_xi    : 0.9987
MAE        : 0.0041
```

The segmentation module's performance was evaluated using the Polyp-PVT model on a set of 20 images. The module-wise performance is summarized in Table (segmentation_metrics) using standard visual recognition evaluation metrics. These include mean Dice coefficient (mDice), mean Intersection over Union (mIoU), weighted F-measure (F_{β^w}), structure measure (S_a), mean enhanced-alignment measure (mE_{ξ}), maximum enhanced-alignment measure ($maxE_{\xi}$), and Mean Absolute Error (MAE).

DETECTION

--- Final Validation Results ---

Model	mAP_50	Precision	Recall	F1_score
YOLOv8	88.54%	86.85%	83.69%	85.24%

The object detection module's performance was evaluated using the YOLOv8 model on the polyp detection dataset obtained from Roboflow. The model was trained and validated using images resized to 640×640 resolution for both the training and validation sets. The performance of the detection module is summarized in Table (detection_metrics) using standard object detection evaluation metrics. These include mean Average Precision at an IoU threshold of 0.5 (mAP@50), precision, recall, and F1-score.

Additional Results and Analysis of Models



We are processing colonoscopy videos using a custom-trained Ultralytics YOLOv8n model with BoT-SORT tracker.

The YOLOv8n model demonstrates strong detection precision (0.8817) and a high AP_{50} (0.7736), indicating that it can localize polyps accurately when overlap thresholds are moderate. However, the lower AP_{mean} (0.5686) and recall (0.6909) suggest that performance drops at stricter IoU thresholds and some polyps are still missed.

Table 1: Detection Performance on PolypGen 2.0

Model	Ultralytics YOLOv8n
Metric	Value
AP_{mean} (mAP@50–95)	0.5686
AP_{50}	0.7736
Precision	0.8817
Recall	0.6909

For tracking, the combination of YOLOv8n with BoT-SORT achieves a high Temporal IoU (0.8863), indicating excellent consistency in maintaining object identity across frames. This shows the pipeline is reliable for video-based polyp tracking despite moderate detection limitations.

Tracking Performance on LD-PolypVideo [15 videos]

Model	YOLOv8n + BoT-SORT
Metric	Value
Temporal IoU	0.8863

Additional Results and Analysis of Models



Segmentation

Done using Polyp-PVT model on Polygen dataset.

PolyPVT achieves moderate segmentation performance on PolypGen, with an mDice of 0.5551 and mIoU of 0.4982, indicating reasonable overlap between predicted and ground-truth masks. The relatively high S_α (0.8231) and E-measure scores (~ 0.86) suggest good structural consistency and boundary alignment. Additionally, the low MAE (0.0783) indicates minimal pixel-wise error, reflecting stable prediction quality.

Metric	Score
mDice	0.5551
mIoU	0.4982
F_β^w	0.5310
S_α	0.8231
mE_ξ	0.8567
$\max E_\xi$	0.8677
MAE	0.0783

VLM

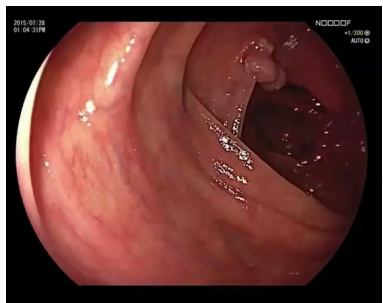
The VLM assistant loads the metadata from the tracking script and lets you chat naturally with Llama-4-Scout (via Groq API) about findings, polyp visuals, stats, comparisons, and clinical advice, and the relevance of questions are verified by Llama-3.1-8B.

Satisfied results were achieved through rigorous QnA with the model about personal colonoscopies and even portraying as doctors to ask for surgical approaches.

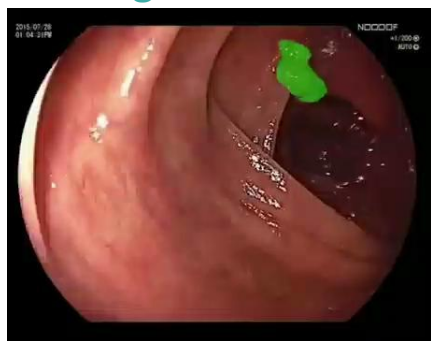
Working Demonstration



Tracking



Segmentation



Final Working Demo [Complete Pipeline]

